

DATA ANALYSIS PROJECT REPORT

Project Title	Sales and Profit Analysis of Superstore Dataset		
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Course / Section	B		
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Date of Submission	02/04/2026		

1. Executive Summary

This project analyses the Superstore dataset containing retail transaction data including sales, profit, discount, category, region, and customer segments. The main objective of this analysis is to identify sales patterns, profitability trends, and relationships between key business variables using data analysis techniques.

The dataset was cleaned by removing duplicates, converting date columns, and handling missing values. New features such as Month and Profit Ratio were engineered to improve analysis depth. Pandas was used for data manipulation, NumPy for numerical calculations, and Matplotlib for visualization.

Key findings show that Technology generates the highest total sales, the South region has comparatively lower sales performance, and there is a clear relationship between discount and profit, where higher discounts often reduce profitability. Additionally, sales peak during November, indicating seasonal trends in consumer behaviour.

These insights can support strategic decisions in pricing, inventory management, and regional marketing.

2. Introduction and Project Objectives

The Superstore dataset contains detailed retail transaction data including information about sales, profit, discount, product categories, customer segments, and regional performance. Each record represents a single order item placed by customers across different locations. This dataset is widely used in data analysis because it helps understand real-world business performance and decision-making patterns in retail operations.

This topic is important because businesses depend on sales and profit analysis to improve performance, identify weak areas, and optimize strategies such as pricing, discounting, and regional targeting. Understanding how different factors affect profit and sales can help companies make data-driven decisions rather than relying on assumptions.

The motivation for this analysis is to explore how different business variables interact with each other and to identify patterns that affect profitability and sales distribution across categories, regions, and time periods.

Item	Content
Dataset name	Superstore Sales Dataset
Dataset source	https://www.kaggle.com/datasets/binib1997/superstore
Main project goal	To analyse sales, profit, and regional performance to identify key business insights and trends using Python-based data analysis
Research Question 1	Which product categories contribute the highest total sales and profit?
Research Question 2	How do sales and profit vary across different regions and months?
Research Question 3 (optional)	How does discount impact profit and overall business performance?

3. Dataset Description

The dataset contains retail transaction records of a superstore. Each row represents a single order line item. It includes information about customers, products, sales, profit, discount, shipping, and geographical regions.

3.1 Basic Dataset Information

Property	Value
Number of rows	9994
Number of columns	21
Data types present	Numerical, Categorical, Date-Time
Target or key variable(s)	Sales, Profit
Potential issues observed at first glance	No missing values but requires type conversion and duplicate handling

3.2 Initial Observations

At first inspection, the dataset was clean with no missing values. However, date columns were in object format and required conversion. The dataset also contains multiple categorical variables such as Region, Category, and Segment which are useful for grouping analysis.

4. Data Cleaning and Preparation

Before performing analysis, the dataset was carefully examined to ensure it was clean, consistent, and suitable for accurate results. Although the Superstore dataset is relatively well-structured, several preprocessing steps were necessary to improve usability for analysis.

First, duplicate records were checked and removed from the dataset. This step was important because duplicate entries can distort results such as total sales, profit calculations, and group-based comparisons. Removing duplicates ensured that each transaction was counted only once, improving data reliability.

Second, the date columns (“Order Date” and “Ship Date”) were converted from text format into proper datetime format. This conversion was necessary because time-based analysis, such as monthly sales trends, cannot be performed accurately when dates are stored as strings. After conversion, it became possible to extract time features such as month and analyse seasonal patterns.

Third, missing values were checked and handled. Although the dataset contained no missing values, a forward-fill method was applied as a precaution to maintain data consistency. This ensured that any potential gaps in the dataset would not interrupt calculations or grouping operations.

Finally, new variables such as “Month” and “Profit Ratio” were created during preparation. These derived features helped in performing deeper analysis, such as understanding monthly sales trends and measuring profitability efficiency across different transactions.

Overall, these cleaning steps ensured that the dataset was structured, consistent, and ready for meaningful exploratory data analysis.

Issue Found	Action Taken	Reason / Justification	Code Reference
Duplicate rows	Removed duplicates	Ensures data accuracy	drop_duplicates()
Date format issue	Converted to datetime	Needed for time-based analysis	pd.to_datetime()
Missing values (if any)	Forward fill used	Maintains continuity	ffill()

5. Feature Engineering

During the analysis, new variables were created from the existing dataset to improve the depth of insights and support the research questions more effectively. These engineered features help transform raw data into more meaningful information that can be used for better interpretation of business performance.

The first engineered feature is **Month**, which was extracted from the “Order Date” column. Since the original dataset only contained full dates, extracting the month allowed for easier analysis of seasonal trends in sales. This feature directly supports the research question related to how sales vary over time, making it possible to identify peak and low-performing months.

The second engineered feature is the **Profit Ratio**, calculated by dividing Profit by Sales. This feature helps measure how efficiently the business is generating profit from its sales. Instead of looking at profit alone, this ratio provides a normalized view of profitability, making it easier to compare performance across different categories, regions, and transactions.

These engineered features were essential because they added analytical depth to the dataset and enabled time-based and efficiency-based analysis, which directly supports the research objectives of the project.

New Feature	How It Was Created	Why It Is Useful
Month	Extracted from Order Date	Used for monthly trend analysis
Profit Ratio	Profit / Sales	Measures profitability efficiency

6. Analysis Methodology

The analysis was conducted using Python with a structured approach to explore, clean, and visualize the Superstore dataset to answer the defined research questions.

Pandas was the primary tool used for data handling and manipulation. It was used to load the dataset, inspect its structure, clean the data, and perform operations such as filtering, grouping, and aggregation. These operations helped in summarizing sales, profit, and category-wise performance across different dimensions such as region, category, and time.

NumPy was used for numerical computations to support deeper statistical analysis. In this project, it was used for calculations such as z-score computation to identify outliers in sales data. This helped in detecting unusual values that significantly deviate from the average sales distribution.

Matplotlib was used for data visualization to better understand patterns and trends in the dataset. Various charts such as bar charts, line graphs, scatter plots, and boxplots were created to visually represent relationships between variables like sales, profit, discount, and time. These visualizations directly supported the research questions by making patterns easier to interpret.

Overall, the combination of Pandas, NumPy, and Matplotlib allowed for a complete workflow of data cleaning, analysis, and visualization to extract meaningful business insights from the dataset.

7. Analysis and Visualizations

7.1 Research Question 1

Which product category generates the highest total sales?

Sales were analysed across different product categories using grouping and aggregation functions in Pandas. The total sales for each category were calculated to compare their performance and identify which category contributes the most revenue.

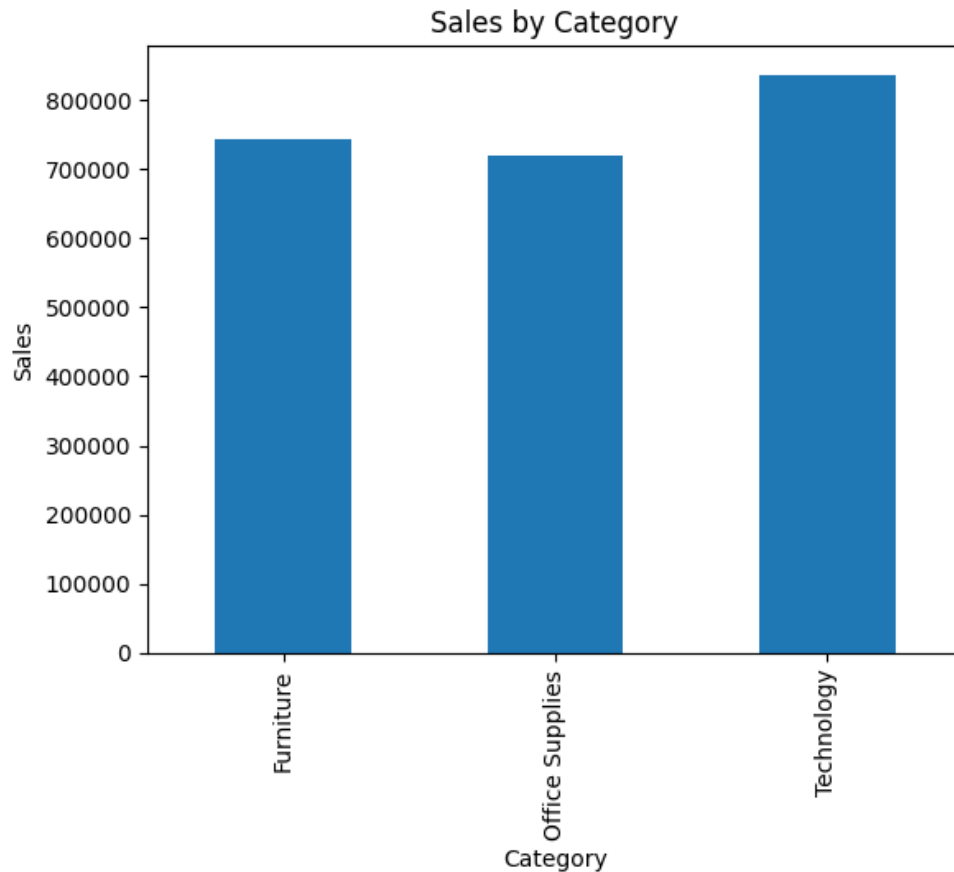


Figure 1. Sales distribution across product categories

The Technology category generates the highest total sales, followed by Furniture and Office Supplies. This indicates that Technology products contribute significantly to overall revenue compared to other categories.

7.2 Research Question 2

How do sales vary across different regions?

Sales performance was compared across different regions by grouping the dataset based on region and calculating total sales for each group.

Region		
Central		501239.8908
East		678781.2400
South		391721.9050
West		725457.8245
Name:	Sales,	dtype: float64

Figure 2. Total sales comparison across different regions

The West and East regions show higher sales compared to the South region. This suggests that the South region has relatively lower market performance and may require improvement strategies.

7.3 Research Question 3

How do sales vary across different months?

Monthly sales trends were analysed by extracting the month from the order date and aggregating total sales for each month.

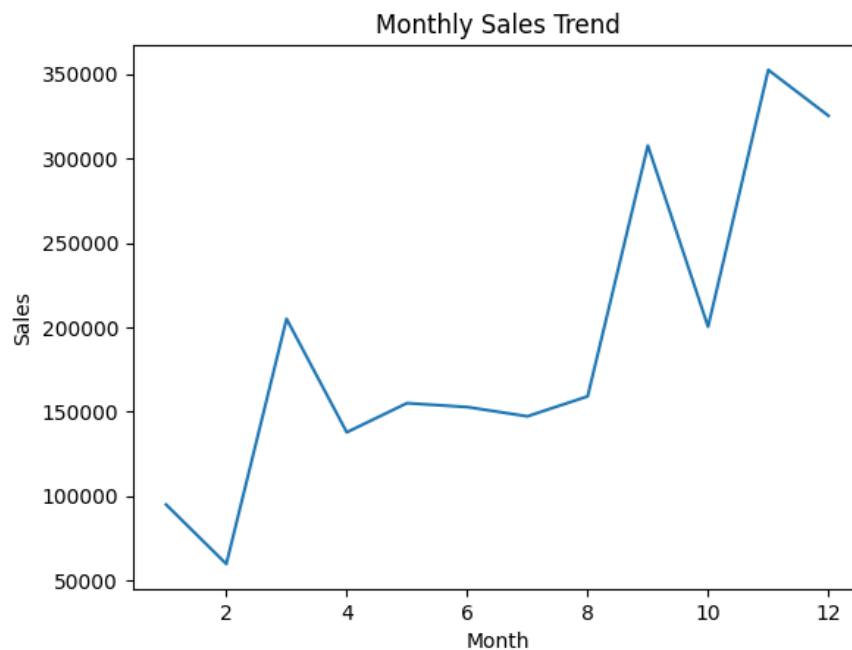


Figure 3. Monthly sales trend across the years

Sales fluctuate throughout the year, with a noticeable peak in November and December. This indicates seasonal buying behaviour, likely influenced by holiday and promotional periods.

7.4 Research Question 4

What is the relationship between sales and profit?

A scatter plot was created to analyse the relationship between sales and profit values in the dataset.

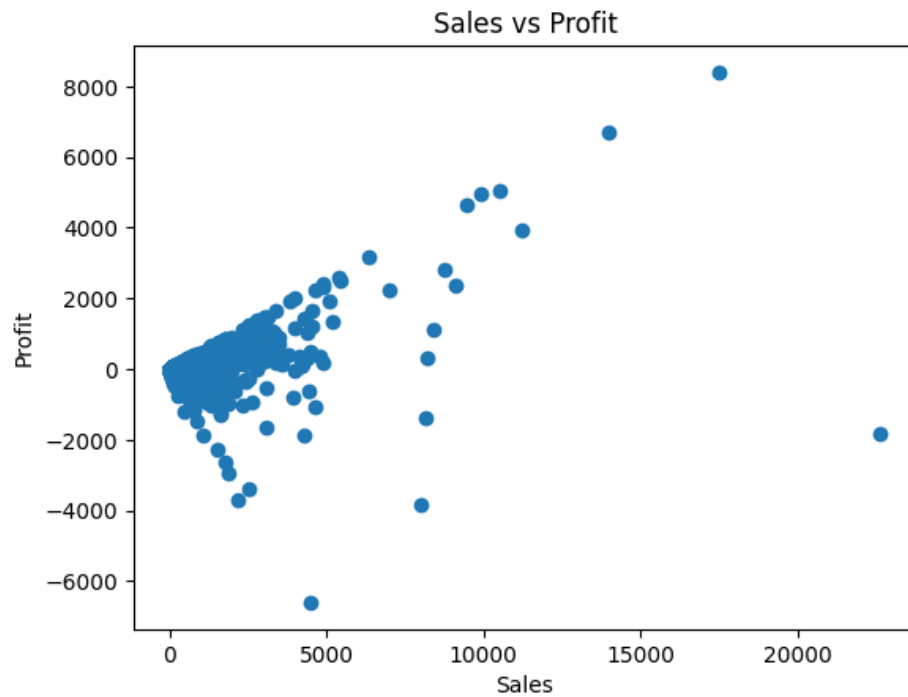


Figure 4. Relationship between sales and profit

The relationship is not strictly linear, as some high sales values correspond to low or negative profit due to heavy discounts.

7.5 Research Question 5

Are there any outliers in the sales data?

A boxplot was used to identify unusual values and outliers in the sales distribution.

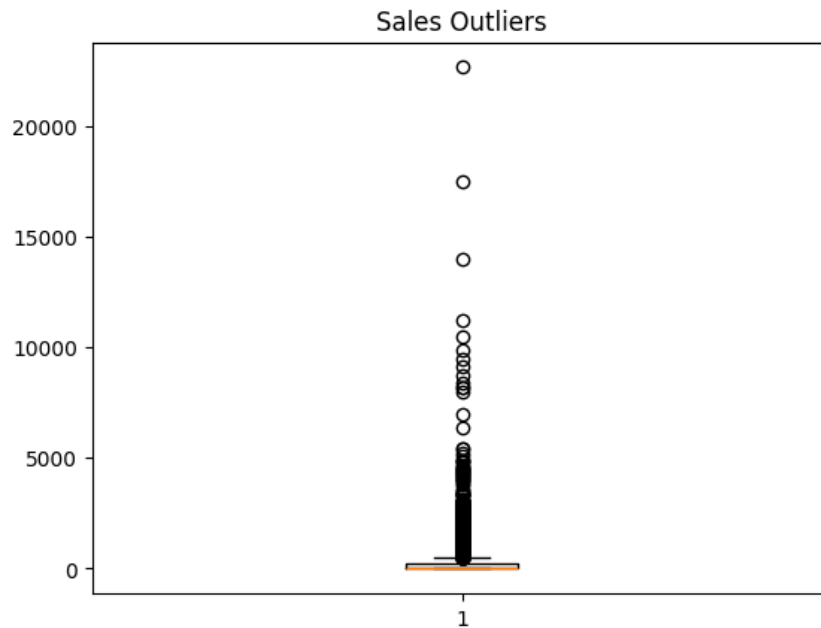


Figure 5. Distribution of sales with outlier detection

The boxplot shows several outliers in sales data, indicating the presence of unusually high-value transactions that differ significantly from the majority of sales.

8. NumPy-Based Custom Computation

NumPy was used to perform a z-score calculation on the Sales data to detect outliers. This method measures how far each value is from the mean in terms of standard deviation, helping identify unusual sales values.

Formula: $Z = (X - \mu) / \sigma$

Where X is the value, μ is the mean, and σ is the standard deviation.

This computation helped identify extreme sales values that significantly differ from the average, improving the understanding of data distribution and variability.

Component	Description
Computation used	Z-score for Sales
Purpose	To detect outliers in sales distribution
Formula or logic	$(x - \text{mean}) / \text{std}$
Result / takeaway	Identified extreme sales values that deviate from normal range

9. Key Findings

1. The Technology category generates the highest total sales among all categories.
2. The South region shows the lowest sales performance compared to other regions.
3. Sales show strong seasonal peaks, especially in November and December.
4. Higher discounts often reduce profit, indicating a negative impact on profitability.
5. Sales data contains significant outliers, representing unusual high-value transactions.
6. Profit ratio varies across categories, indicating differences in profitability efficiency.

10. Limitations

- The dataset does not include customer demographic or behavioural information.
- External factors such as marketing campaigns or economic conditions are not included.
- The analysis is based on historical data and cannot guarantee future predictions.
- Correlation between variables does not necessarily imply causation.

11. Conclusion

This project successfully analysed the Superstore dataset using Python tools to extract meaningful business insights. The analysis revealed that Technology is the most profitable category, while the South region underperforms in sales. Seasonal trends were also observed, with peak sales occurring in the last quarter of the year.

Additionally, the study found that discounts can negatively affect profit margins, and several outliers exist in the sales data. These insights can help businesses improve decision-making and optimize sales strategies. Overall, the analysis demonstrates how data-driven insights can support better business decision-making.

12. References / Dataset Source

Dataset Source:

<https://www.kaggle.com/datasets/binib1997/superstore>

Project Repository:

<https://github.com/FatimaAmeerFariha/superstore-data-analysis>